INCH-POUND

T-T-616C
14 April 2026
SUPERSEDING
T-T-616B
12 June 1986

FEDERAL SPECIFICATION

TREATMENT: MILDEW RESISTANT, FOR ROPE AND CORD

Inactive for New Design after 3 December 1998

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of mildew-resistant treatment of rope and cord.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL STANDARD

FED-STD-191 - Textile Test Methods

(Copies of these documents are available online at https://assist.dla.mil.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless an issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4020

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2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes

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applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rope and cord. The specification governing the rope and cord to be treated under this specification shall be cited in the applicable procurement document (see 6.2). Except as otherwise indicated herein, rope and cord shall conform to the requirements of the applicable specification for the end item prior to and after the mildew-resistant treatment.

3.2 Treatment. The mildew inhibitor shall be copper-8-quinolinolate. The use of any other inhibitor is prohibited. Any petroleum fractions, oils or resins used shall be of the nonoxidizing type and shall be free of naphthenic acids or its salts. The inhibitor shall be thoroughly dispersed and so formulated that it may be applied to effect proper penetration into the rope or cord.

3.2.1 Application (hard and soft fiber ropes – manila, sisal, hemp, or jute). For hard and soft fiber ropes, other than cotton, the copper-8-quinolinolate shall be applied to the fibers, yarns, or strands prior to twisting (laying) into the finished rope. The treatment shall be uniformly applied throughout the entire cross section of the rope.

3.2.1.1 Cotton ropes and hard and soft fiber cords. The copper-8-quinolinolate shall be applied to the rope or cord as an after treatment. The copper-8-quinolinolate may be applied to the rope or cord with a water-repellent treatment, when required in the applicable end item specification.

3.2.2 Concentration. The application of the inhibitor shall deposit within the rope or cord a minimum of 0.13 percent to a maximum of 0.40 percent copper as metal from copper-8-quinolinolate when tested as specified in 4.2.4.

3.2.3 Prohibited materials. The use of casein, glue, gum, starch, dextrin, water-soluble materials, paint dryers, resin or vegetable oils, oxidizing oils or resin modified with such oils is prohibited except where required for use in polished finishes on cord.

3.3 Color. Unless otherwise specified in the applicable end-item specification or procurement document, the color of the treated rope or cord shall be the natural color resulting from the treatment (see 6.3).

3.4 Acidity (pH) after aging. The mildew-resistant treated rope or cord shall have a pH value of not less than 5.0 before and after heat aging when tested as specified in 4.2.4.

3.5 Breaking strength. A 10 percent loss in breaking strength, based on the minimum specified for the untreated rope or cord, will be permitted, due to the application of the treatment when tested as specified in 4.2.4.

3.6 Handling characteristics. The inherent handling qualities of the untreated ropes or cords shall not be altered by the application of the mildew-resistant treatment. The mildew-resistant treated finish on the ropes or cords shall not crack, peel, dust or flake from the treated surfaces.

3.7 Workmanship. The treated rope or cord shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ-Z1.4.

4.2.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item visual examination. The end item shall be examined in accordance with the applicable classification of defects for the untreated material and as listed in table I. In the event that a condition listed in table I represents a duplication of a condition cited in another applicable document, the condition shall be counted only once. The defects listed in table I shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one packaged unit of rope or cord. Ten percent of the length contained in each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in packaged units of rope or cord. The inspection level shall be level I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE I. End item visual defects.

Examine	Defects	Classification	
		Major	Minor
Appearance and workmanship	Cut (any)	X	
	Chafed or damaged	X	
	Finish other than specified	X	
	Finish cracked, peeled, dusted, or flaked	X	
	Lumps		X
Cleanliness	Spot or stain clearly visible 1/		X
	Overall uncleanliness	X	
Color	Other than natural (unless otherwise specified)	X	
	Specific shade specified not within established tolerances		X

1/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for weight or length and winding. These examinations shall be conducted in accordance with the provisions of the applicable specification for the untreated material or as specified in the contract or order.

4.2.4 End item testing. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table III, shall be followed. When the "Number of determinations" and "Results reported as" are not specified in table III, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table II. The sample unit for testing shall be one packaged unit of rope or cord. The lot size shall be expressed in packaged units of rope or cord. All test reports shall contain the individual values utilized in expressing the final result. The lot shall

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be unacceptable if one or more sample units fail to meet any requirement. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions, except in settlement of dispute in which case the test the test should be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE II. Sampling for tests.

Lot size (No. of packaged units of rope or cord)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item tests.

Characteristic	Requirement paragraph	Test method	No. of determinations per individual sample unit	Results reported as
Treatment:				
Inhibitor and compounds present	3.2	1/ 2/	-	-
Application to:				
Hard and soft fiber ropes	3.2.1	1/	-	-
Cotton ropes and hard and soft fiber cords	3.2.1.1	1/	-	-
Cross-sectional uniformity	3.2.1	Visual	1	Pass or fail
Concentration of copper-8-quinolinolate (percent copper)	3.2.2	2050.1	-	-
Prohibited materials	3.2.3	1/	-	-
Acidity (pH) after aging	3.4	4.2.4.1	2	-
Breaking strength	3.5	3/	1	Reported to the nearest 0.1 percent

1/ "Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement."

2/ The certificate of compliance for the requirements as specified in 3.2 shall incorporate a statement that specific inhibitor (copper-8-quinolinolate) was used in the treatment of the rope or cord and that any petroleum fractions, oils or resins used are of the nonoxidizing type and are free of naphthenic acid or its salts.

3/ Breaking strength shall be determined in conformance with the method specified in the applicable specification for the untreated material.

4.2.4.1 Acidity (pH) before and after heat aging.

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a. The pH of the mildew resistant treated rope, yarns or cord shall be tested prior to aging in accordance with method 2811 of FED-STD-191.

b. pH after aging. Samples (approximately 10 grams) shall be put in weighing bottles (uncovered at a temperature of 175° to 180°F) for 44 hours. At the end of this period the yarns or cord shall be tested for acidity (pH) in accordance with method 2811 of FED-STD-191.

5. PACKAGING. (Not applicable)

6. NOTES
(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. The mildew-resistant treatment is intended for use with ropes and cords.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of specification governing the untreated rope or cord.
c. Color, if other than required (see 3.3 and 6.3).

6.3 Standard sample. If a specific shade (color) is specified and reference is made to a standard sample, for access to that standard sample address the procuring office issuing the invitation for bids.

6.4 Subject term (key word) listing.

Copper-8-quinolinolate
Manila
Sisal
Hemp
Jute

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - AS
Air Force - 71
DLA - CQ

Preparing activity:
DLA - CQ

(Project 4020-2026-005)

Review activities:
Army - MD, MI
Navy - MC

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